PARTS LIST

Semiconductors:	
Board1	- Arduino board
D1	- 1N4007 rectifier diode
Resistors (all 1/4-watt, ±5% carbon):	
R1	- 10-kilo-ohm
R2-R3	- 1-kilo-ohm
T1, T2	- BC547 NPN transistor
Miscellaneous:	
CON1, CON3, CON4	- 2-pin connector
CON2	- 3-pin connector for PIR
PZ1	- Piezo buzzer
	- PIR motion sensor
	- LDR
	- Breadboard
	- Jumper wires
	- 12V, 1C/O relay
	- 60W, 20V AC bulb

sensor detects motion by sensing changes in ambient temperature when a human body passes by, effectively controlling the switch when motion is detected. The PIR module includes two variable resistors, one for time setting and the other for range setting. You can adjust these presets to set the minimum time and maximum range before using them in the mechanism.

During the night, this security light is enabled because of the LDR used in the project. The LDR (light dependent resistor) measures the light level. At night, the voltage at A1 (pin10) of the Arduino Uno board will be high due to the high resistance of the LDR, enabling the circuit. In contrast, during the day, the voltage at A1 will be low due to the low resistance of the LDR, disabling the circuit.

So, this night security system is triggered when a logic high-level signal is detected at its sensor input port (A1 of Arduino Uno board). Only one output (D6 pin21 of Arduino Uno board) is used, which is connected to both transistors T1 and T2.

The operation of the circuit is straightforward. At night, when an intruder comes within the of the PIR sensor, the logic input at

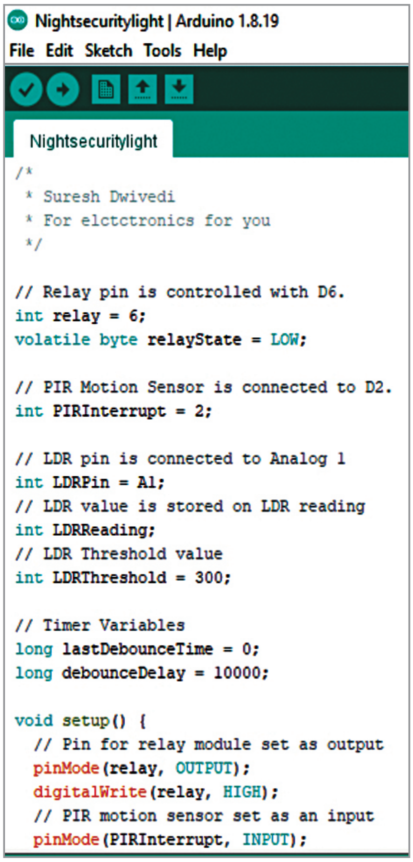

Fig. 3: Screen shot of the source code

pin A1 goes high. In this condition, output D6 is also high, which makes both transistors T1 and T2 conductive. Transistor T1 drives the relay, and transistor T2 drives the piezo buzzer.

As a result, relay RL1 is energised, turning on the light by providing 230V AC via its contacts.

EFY Note
The source code of this project is available for free download at *www.electronicsforu.com*

Simultaneously, the piezo buzzer sounds. Both the buzzer and the light remain enabled for ten seconds to alert security at the gate.

Software

The logical software is written in Arduino IDE version 1.8.5. Before uploading the given sketch to the Arduino Uno board, ensure that you select the correct board. After selecting the port, upload the sketch file 'Nightsecuritylight.ino' to the board. Fig. 3 shows a screenshot of the source code.

Construction and testing

To upload the source code, use a USB cable to connect the Arduino board to a laptop/desktop. After uploading the source code, remove the USB cable and connect a 12V DC battery to the board to power the project.

Next, you can assemble the circuit based on the circuit diagram and enclose it in a suitable small box, including the Arduino Uno board. Position all LDRs at the top of the box so that they receive light during the day. Ensure that the bulb's light does not fall on the LDR. Attach the bulb and buzzer to the rear side of the cabinet. You can extend the bulb using twin wires to a suitable location as desired. Also, install the PIR sensor on one side of the cabinet so that it can capture/detect a person's movement. Instead of a 12V battery, you can use a 12V power adaptor at the DC terminal of the Arduino Uno board.

Bonus. You can watch the video of the tutorial of this DIY project at *https://www.electronicsforu.com/videos-slideshows/live-diy-Arduino-based-securitylight* EFY

S.C. Dwivedi is an electronics enthusiast and circuit designer at EFY